

Introduction to MongoDB



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What You will learn?

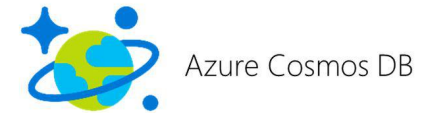
- What is NoSQL Database?
- Types of NoSQL Database
- Introduction to MongoDB
- Why MongoDB ?
- MongoDB Advantages and Limitations
- MongoDB Installation and Configuration
- MongoDB: Hierarchical Objects
- Mongo Shell

What is NoSQL Database?

- NoSQL stands for Not Only SQL
- Schema less
- Avoid Joins
- Supports Horizontal and Vertical Scaling
- Non-locking concurrency control mechanism
- Provide higher performance per node than RDBMS

Types of NoSQL Database

- Document-oriented



- Key-Value



- Big Table/Column families



- Graph Store



Introduction to MongoDB

- Introduced in 2009 by 10gen, known as MongoDB Inc.
- NoSQL database written in C, C++ and JavaScript
- Document-oriented database
- Supports dynamic schema – No DDL
- Stores data in JSON/BSON format
- Supports multiple platform like Windows, Linux, Mac etc.
- Open-source

Why MongoDB?

- MongoDB stores data in objects (JSON, BSON).
- Now-a-days, programmers write code in object oriented fashion using languages like C#, Python, Php, Java etc.
- Hence, programmers need a database which can store the data in objects.
- Since, querying and manipulating data objects is easy and it reduces the time of database operations.
- Embedded documents and arrays reduce need of joins.

MongoDB Features

- Document-Oriented storage
- Full Index Support
- Replication & High
- Availability
- Auto-Sharding
- Querying
- Fast In-Place Updates
- Map/Reduce functionality

Who Use MongoDB?



Advantages of MongoDB

- Supports APIs (drivers) for many languages like C++, C#, Java, JavaScript, Python, Ruby, Perl, Java Scala etc.
- Supports Caching for speed.
- Supports Replication and Failover for High Availability.
- Supports Auto Sharding for Easy Scalability.
- Supports Indexes for high performance.
- Supports Map/Reduce for Aggregation.

What MongoDB doesn't have?

- DDL
- Joins
- Transactions
- Procedures
- Triggers

Setting up MongoDB

Select the server you would like to run:

MongoDB Community Server

FEATURE RICH. DEVELOPER READY.

MongoDB Enterprise Server

ADVANCED FEATURES. PERFORMANCE GRADE.

Version

4.2.2 (current release)

OS

Windows x64 x64

Package

MSI

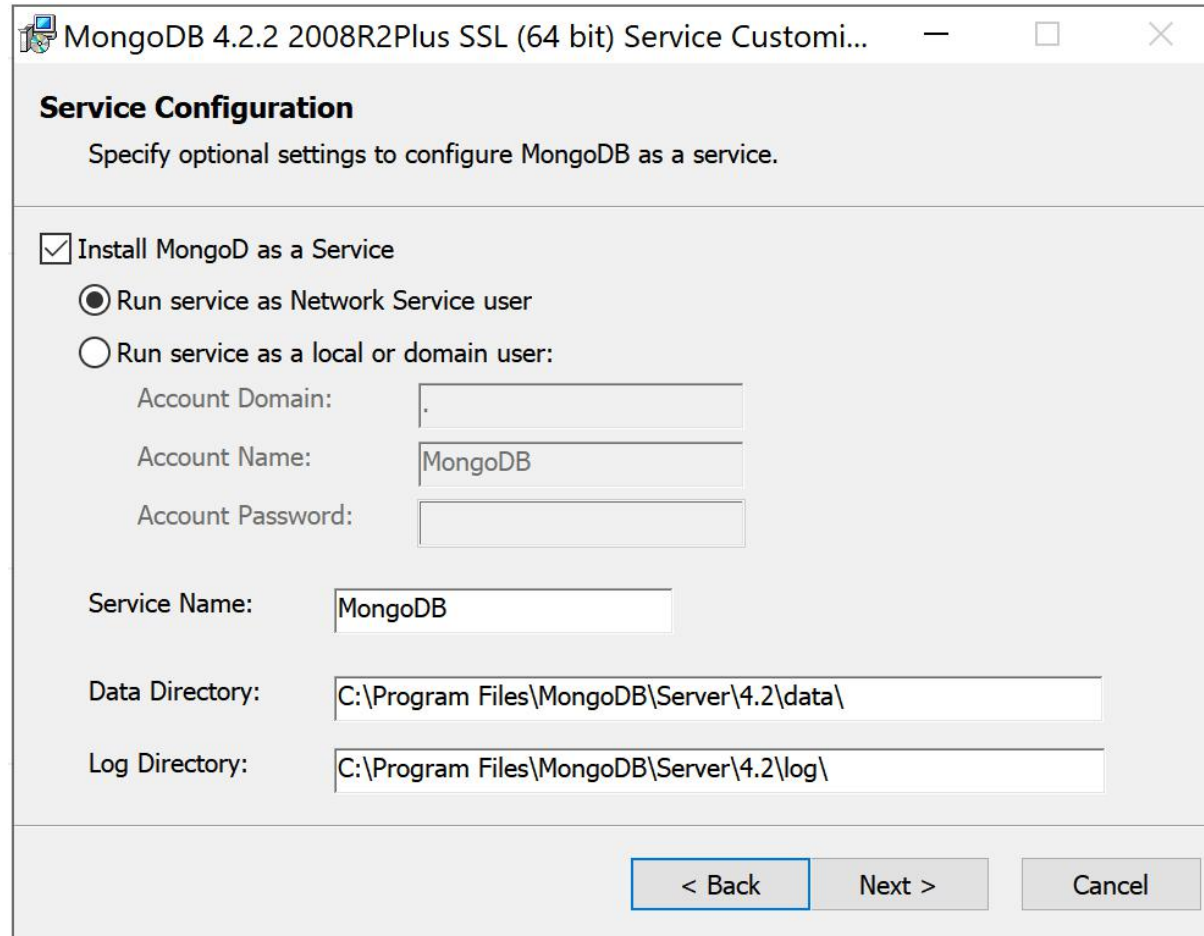
Download

https://fastdl.mongodb.org/win32/mongodb-win32-x86_64-2012plus-4.2.2-signed.msi

<https://www.mongodb.com/download-center/community>

- Release notes
- Changelog
- All version binaries
- Installation instructions
- Download source (tgz)
- Download source (zip)

MongoDB Configuration



The screenshot shows the 'Service Configuration' window for MongoDB 4.2.2. The window title is 'MongoDB 4.2.2 2008R2Plus SSL (64 bit) Service Customi...'. The main heading is 'Service Configuration' with the instruction 'Specify optional settings to configure MongoDB as a service.'.

The 'Install MongoDB as a Service' checkbox is checked. Below it, there are two radio button options: 'Run service as Network Service user' (selected) and 'Run service as a local or domain user:'. Under the second option, there are three text boxes: 'Account Domain:' (containing a period), 'Account Name:' (containing 'MongoDB'), and 'Account Password:' (empty).

Below these are three more text boxes: 'Service Name:' (containing 'MongoDB'), 'Data Directory:' (containing 'C:\Program Files\MongoDB\Server\4.2\data\'), and 'Log Directory:' (containing 'C:\Program Files\MongoDB\Server\4.2\log\').

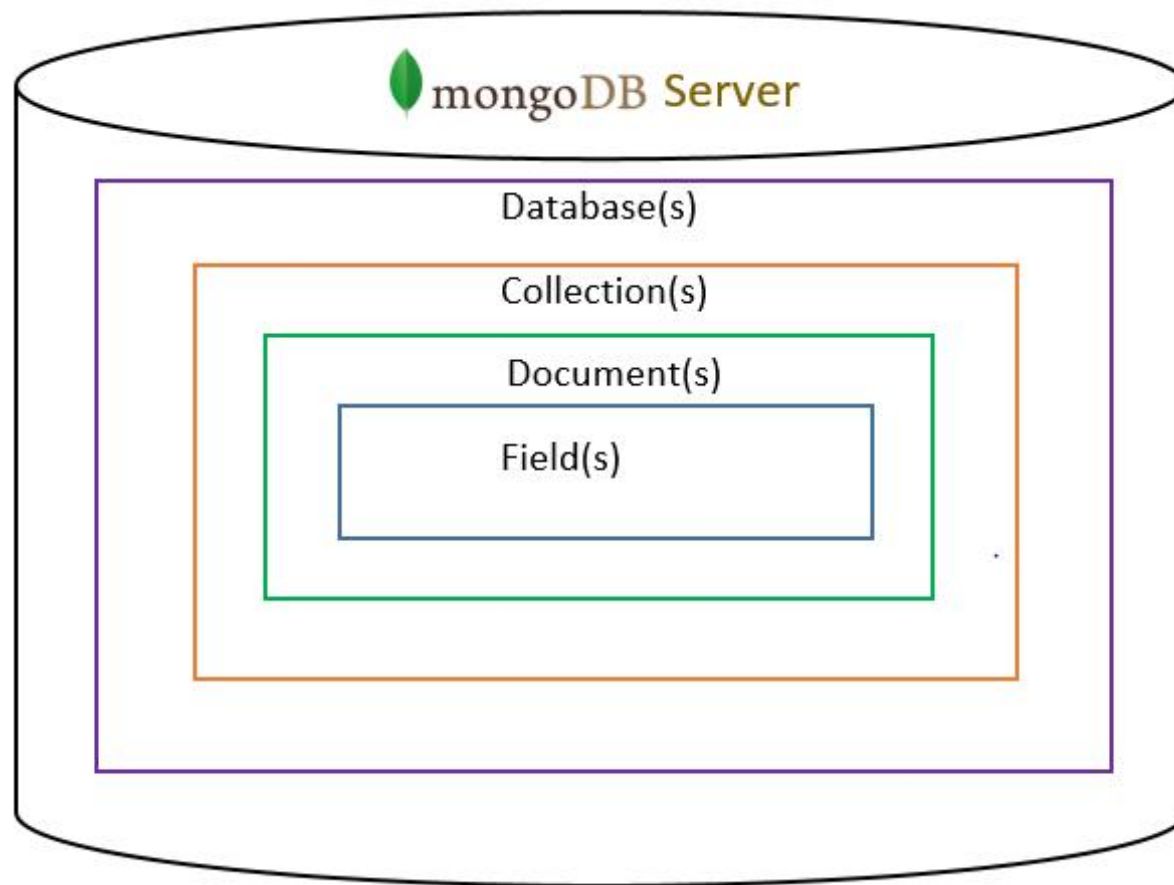
At the bottom right, there are three buttons: '< Back' (highlighted with a blue border), 'Next >', and 'Cancel'.

Verify MongoDB Installation

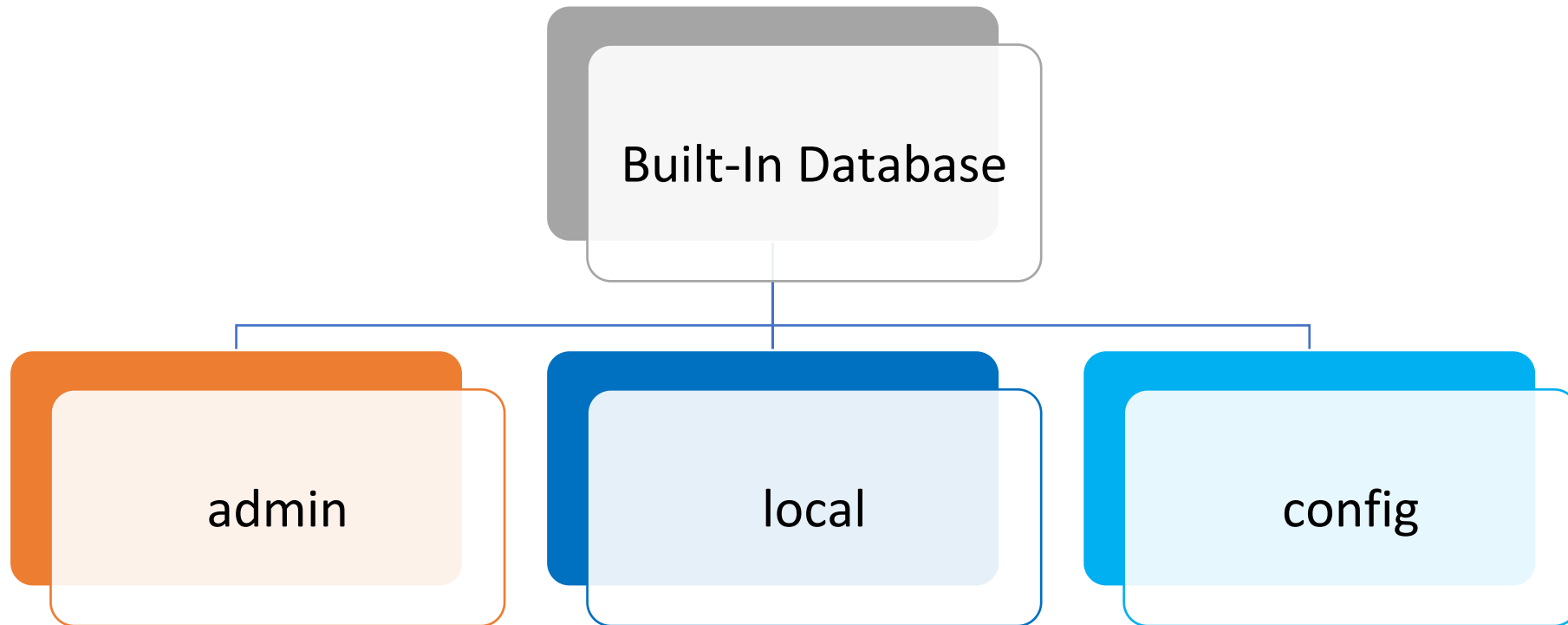
RDBMS Concepts to NoSQL

RDBMS	NO SQL
Database	Database
Table, View	Collection
Row	Document
Column	Field
Primary Key	_id field
Foreign Key	References
Index	Index
Join	Embedded Document
Partition	Shard

MongoDB: Hierarchical Objects



Built-In Databases



User defined Database

- Created on-the-fly when referenced first time.
- Using MongoDB Compass can be created manually.
- Contains Multiple Collections.

Collection

- Created on-the-fly when referenced first time.
- Schema-less.
- Contains Documents.
- Supports Indexes based on one or more fields.

Document

- Created on-the-fly when referenced first time.
- Has `_id` field which acts as Primary key in RDBMS.
- Supports Relationships using Embedded or References.
- Stores data in BSON (Binary form of JSON).

Mongo Shell

- Mongo Shell is an interactive JavaScript Interface for MongoDB.
- Can run any type of query or command related to MongoDB
- To Start Mongo Shell, run command `mongo` after setting up environment variable for `mongodb`.

Mongo Shell Basic Commands

- For List of Database → `show dbs`
- For connect any database → `use <database>`
- For List of Collections in Database → `show collections`
- For create database → `use <database>`
- For create collections → `db.createCollection(collectionname)`

Learn. Build. Empower.